



# MAR BASELIOS INSTITUTE OF TECHNOLOGY AND SCIENCE NELLIMATTOM, KOTHAMANGALAM



## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING KNOTBOOK AN ONLINE LEARNING PLATFORM

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# ABSTRACT

- Knotbook is an online platform that facilitates easy sharing of educational notes
- Automatically creates various practice questions (multiple-choice, true/false, short-answer) from uploaded notes using NLP
- The system generates a comprehensive collection of assessments without manual input
- Provides an interactive learning experience that boosts student involvement
- Data analysis helps to forecast student performance and identify at-risk learners

# INTRODUCTION

- Online learning has evolved rapidly due to advancements in Internet and communication technology
- Despite benefits, challenges like low engagement and high dropout rates remain
- A Virtual Learning Environment is needed to enhance educational experience and improve student engagement
- VLE automates assessment and monitoring by generating diverse questions
- This system has potential to transform education by improving student outcomes, increasing retention rates, and enabling more personalized learning experiences

# OBJECTIVES

- To create a platform for faculty to upload, organize, and share lecture notes with students
- To enable students to take online exams with system-generated questions from uploaded notes
- To allow students to use text-voice conversion facility during exams
- To use analytics to identify students at risk and notify the appropriate users

# LITERATURE SURVEY

## 1. MCQGen: A Large Language Model-Driven MCQ Generator for Personalized Learning [1]

### Methodology:

- Large Language Model (LLM) Integration: GPT-4
- Retrieval-Augmented Generation (RAG): LLM- external knowledge base
- Prompt Engineering Techniques

Chain-of-Thought (CoT): Guides LLM to generate questions

Self-Refine: LLM evaluates and improves its own outputs

- Feedback Loop: Crowd-sourced feedback mechanism

# LITERATURE SURVEY

## Advantages

- Feedback Mechanism
- Guidance for LLM

## Disadvantages

- Limitation in Math-Based Questions
- Question Diversity and Complexity
- Dependence on Instructor Input
- Limited Subject Domains

# LITERATURE SURVEY

## 2.UniRaG: Unification, Retrieval, and Generation for Multimodal Question Answering With Pre-Trained Language Models [2]

### Methodology:

- Unified Knowledge Representation:
  - Image to Text: LLaVA (Large Language and Vision Assistant)
  - Table to Text
- Context Retrieval: A cross-encoder model
- Answer Generation: Pre-trained language model (Flan-T5-Base)



# LITERATURE SURVEY

## Advantages

- High-Quality Answer Generation
- Detailed Image Descriptions
- Preserved Tabular Information

## Disadvantages

- High Computational Complexity
- Potential Information Overload
- Risk of Conversion Errors

# LITERATURE SURVEY

## 3. Opinerium: Subjective Question Generation Using Large Language

### Models [3]

#### Methodology:

- Dataset: Set of 40,000 news articles and questions
- Model Training:

Flan-T5: Trained on their own computers.

GPT-3: Trained on OpenAI's servers.

- Testing and Comparing: GPT-3.5
- Evaluation

# LITERATURE SURVEY

## Advantages

- Emphasis on Personal Insights
- Enhanced Performance through Fine-Tuning
- Rigorous Quality Evaluation

## Disadvantages

- Challenging Quality Assessment
- Significant Computing Costs
- Restricted Model Access

# LITERATURE SURVEY

## 4. Leveraging Structured Information from a Passage to Generate Questions [4]

### Methodology:

- Dataset Construction: RACE4QG
- Model Architecture:
  - Encoder: GRU is used and GAT embedding as input.
  - Decoder: Employs attention and pointer networks
  - GAT (Graph Attention Network): Enriches word embeddings

# LITERATURE SURVEY

## Advantages

- Multi-source Input
- Template-based and Neural Techniques Integration

## Disadvantages

- Integration with Unstructured Data May Be Challenging
- Higher Computational Overhead
- Limited Question Type Suitability

# LITERATURE SURVEY

## 5. A study of an Intelligent Algorithm for Automatic Test Paper Generation and scoring in University English Exams [5]

### Methodology:

- Test Paper Generation Model: To generate test papers meeting constraints like difficulty, score, coverage, and timing.
- Improved Genetic Algorithm (IGA):
  - Initialization
  - Fitness Function
- Automatic Scoring for Native-to-English Translation: Evaluating both grammar and semantic similarity.

# LITERATURE SURVEY

## Advantages

- Balanced Question Distribution
- Customizable Difficulty Levels
- Automated Scoring

## Disadvantages

- Limited Scoring for Subjective Responses
- Complexity of Genetic Algorithm
- Dependence on Initial Constraints

# LITERATURE SURVEY

## 6. Automatic Exam Correction Framework (AECF) for the MCQs, Essays, and Equations Matching [6]

### Methodology:

- Word Embedding Models: Word2Vec, FastText, Glove, and USE used to convert text into numerical vectors.
- Equation Similarity Algorithm: HMB-MMS-EMA algorithm.
- Cosine and Pearson Similarity: Used to evaluate the similarity between word vectors and expressions.
- Python Libraries: Gensim, SpaCy, NLTK and SymPy



# LITERATURE SURVEY

## Advantages

- Hybrid Question Type Handling
- Advanced NLP for Essay Evaluation
- Intelligent Equation Matching

## Disadvantages

- Limited Equation Functions
- High Computational Requirements
- Manual Refinement Still Needed
- Restricted to Specific Dataset

# LITERATURE SURVEY

## 7.Toward Robust Evaluation: A Comprehensive Taxonomy of Datasets and Metrics for Open Domain Question Answering in the Era of Large Language Models [7]

### Methodology:

- ODQA Approaches:
  - Retriever-Reader
  - Retriever-Only
  - Generator-Only
- Datasets and Metrics
- LLM-based Evaluation

# LITERATURE SURVEY

## Advantages

- Comprehensive Evaluation Taxonomy
- Holistic Approach to Evaluation
- Tailored for Large Language Models

## Disadvantages

- Focus on Large Language Models Might Exclude Smaller Models
- High Complexity of Evaluation Metrics:
- Limited Focus on Real-Time Applications

# LITERATURE SURVEY

## 8. Automatic Question Generation from Given Paragraph ,IJSRD [8]

### Methodology:

- Text Preprocessing: Tokenizes sentences, words, and performs POS tagging.
- Question Generation Algorithm: Uses Seq2Seq, Transformers.
- Entity Recognition: Identifies key entities and phrases for question focus.
- Template Matching: Matches sentences with predefined question templates.
- Evaluation: Refines questions using metrics like BLEU or human feedback.

# LITERATURE SURVEY

## **Advantages**

- Personalized Learning
- Broad Applicability Across Domains
- Promotes Reading Comprehension

## **Disadvantages**

- Dependence on Structured Input
- Inability to Handle Ambiguity
- Limited Scope of Question Types

# LITERATURE SURVEY

## 9. Investigating Educational and Noneducational Answer Selection for Educational Question Generation [9]

### Methodology:

- Data Preprocessing: Tokenization and POS tagging, using NER for educational entities.
- Answer Selection Model: Classifiers (SVM, Random Forest) or neural networks (BERT)
- Question Generation Model: Seq2Seq or Transformer models
- Semantic Analysis: Word embeddings , cosine similarity
- Evaluation Metrics: BLEU and ROUGE.

# LITERATURE SURVEY

## Advantages

- Dual Mode Answer Selection
- Contextual Relevance for Answer Selection
- Teacher-Driven Customization

## Disadvantages

- Potential for Lower Accuracy in Question Relevance
- Difficulty in Handling Complex Domains
- Challenges in Distinguishing Educational from Non Educational Content

# LITERATURE SURVEY

## 10. A Systematic Review of Automatic Question Generation for Educational Purposes [10]

### Methodology:

- Data Collection: Gather studies on educational question generation.
- Classification: Categorize methods (template-based, rule-based, ML models).
- Model Implementation: Use NLP pipelines and neural networks.
- Evaluation: Metrics like BLEU, ROUGE, and human assessments.
- Findings: Trends toward deep learning and hybrid approaches.



# LITERATURE SURVEY

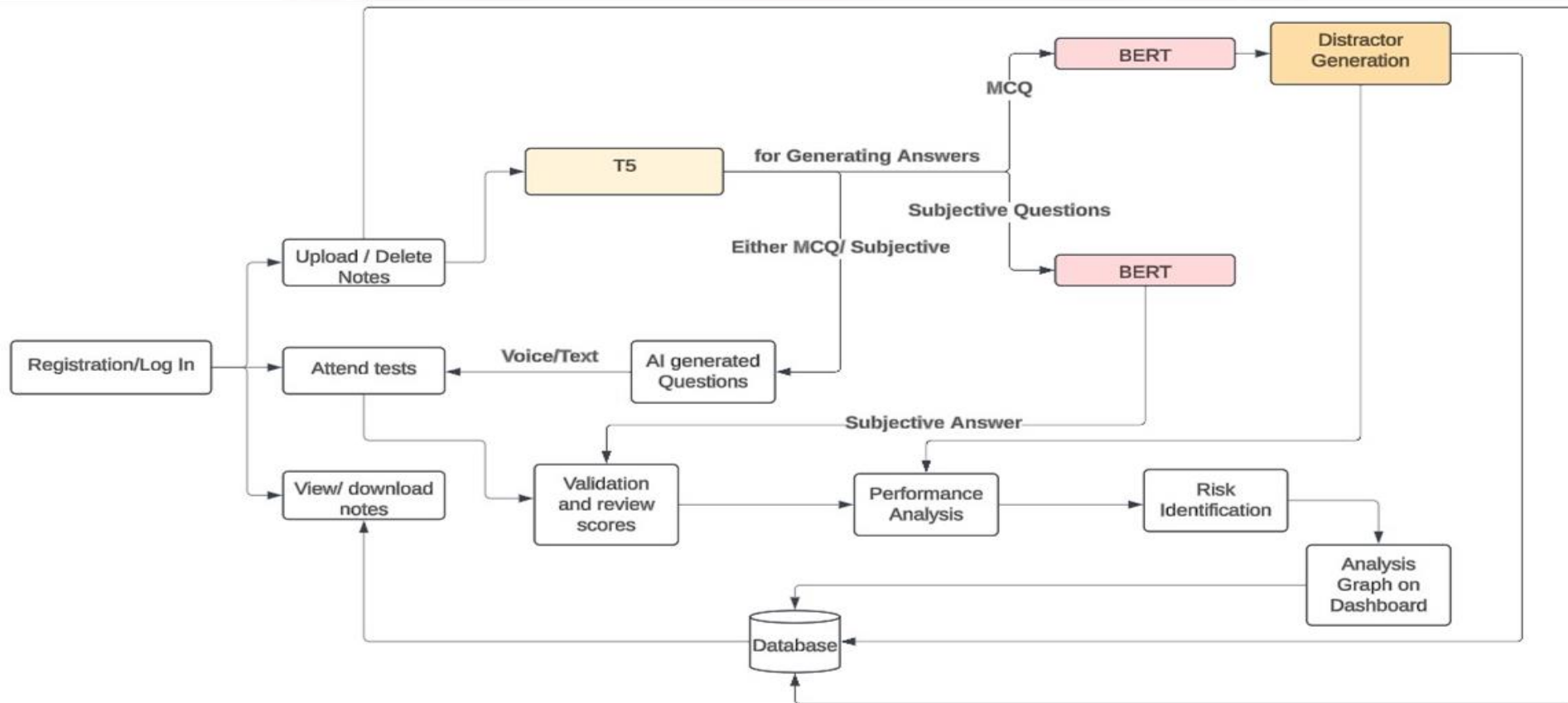
## Advantages

- Comprehensive Review Across Multiple Domains
- Focus on Evaluation and Benchmarks
- Multilingual and Multicultural Capabilities

## Disadvantages

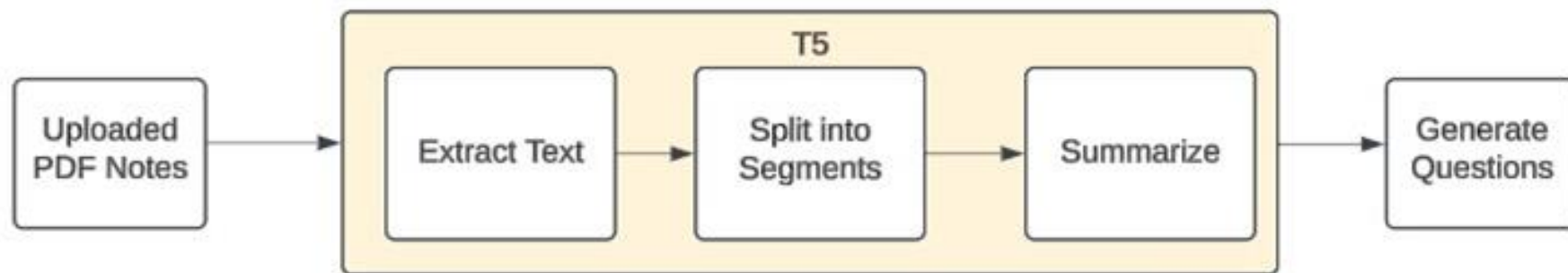
- Limited Insights into Advanced AI capabilities
- Training data dependency
- Bias from Available Literature

# ARCHITECTURE DESIGN



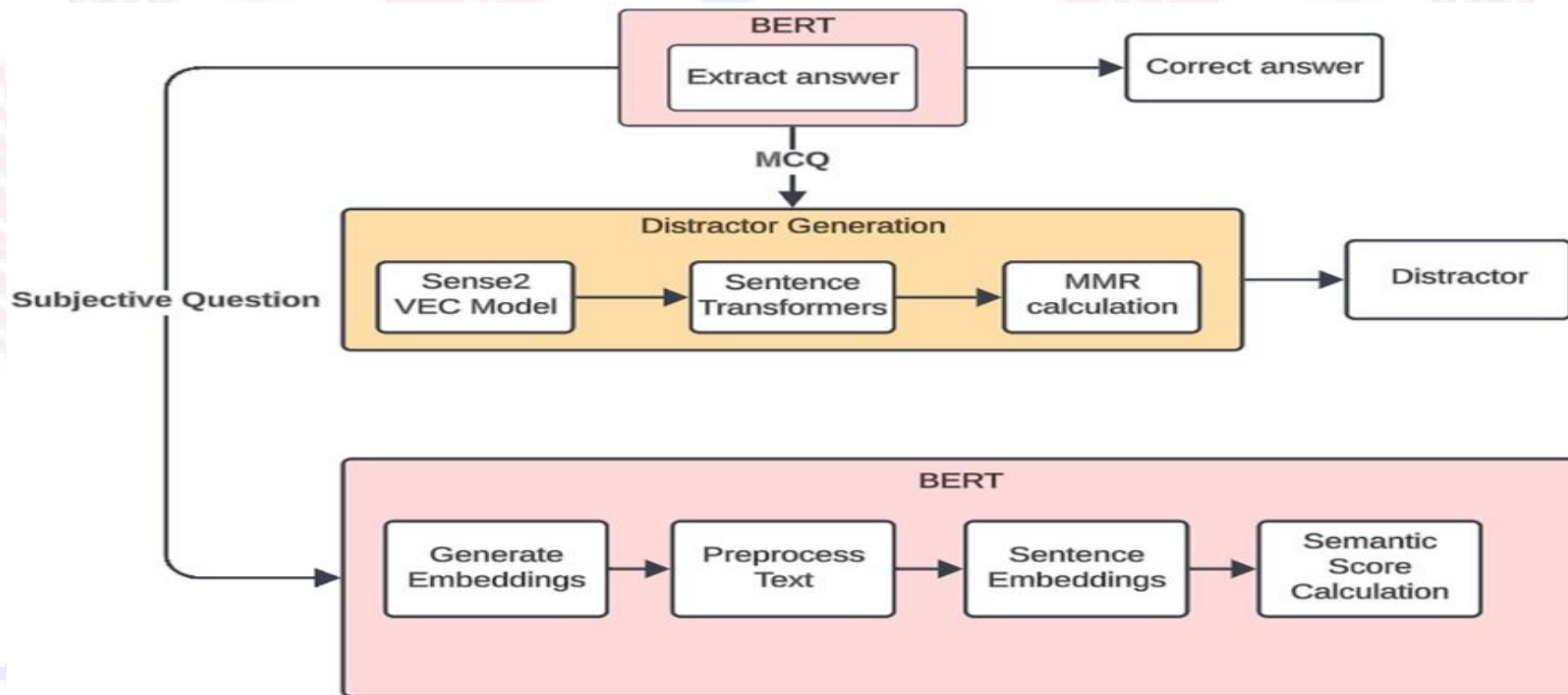
# ARCHITECTURE DESIGN

## T5-Model



# ARCHITECTURE DESIGN

## BERT-Model



# MODULE DETAILS

- Automatic Question Answer Generation
- Text-Voice Conversion
- Answer Validation
- Progress Reporting

# MODULE DETAILS

## **Automatic Question Answer Generation:**

- Question Types: Multiple-choice, true/false, short answer.
- Text Extraction: Extract text content from PDF notes.
- Question and Answer Generation: Using T5 Transformer model.

## **Text-Voice Conversion:**

- Text-to-Voice Engine: Integration with text-to-speech (TTS) services.
- Voice-to-Text Engine: Integration with speech-to-text (STT) services.

# MODULE DETAILS

## **Answer Validation:**

- Automatic Grading: Validate and score student answers.
- Grading System: Assign scores based on correctness.

## **Progress Reporting:**

- Student Performance Tracking: Track and display student progress.
- Performance Analytics: Detailed analysis of student performance over time.
- Progress Dashboard: Visualize student performance metrics.

# IMPLEMENTATION

## PDF Upload and Text Extraction

- Text Extraction: PyPDF2 to extract text from the uploaded PDF document.
- Segmenting Text: Split the extracted text into manageable segments or paragraphs

## Text Summarization

- Load T5-small Model: Use the T5-small model for text summarization.
- Summarization Process: For each segment of text, feed it into the T5-small model to generate a concise summary.



# IMPLEMENTATION

## Question Generation

- Load T5-base Model: Use the T5-base model specifically for generating questions.
- MCQ and Subjective Question Generation

## Answer Extraction for Questions

- BERT Integration: Use BERT model to extract the correct answer from the summarized text.
- Storing Questions and Answers: Store the generated questions and their corresponding correct answers in a database

# IMPLEMENTATION

## **Distractor Generation for MCQs**

- Sense2Vec model: To find semantically similar words or phrases.
- Sentence Transformers: Generate Embeddings for Semantic Similarity
- Maximal Marginal Relevance (MMR): Apply MMR to select distractors that are contextually relevant but distinct from correct answer
- Final MCQ Options: Combine the correct answer and generated distractors to create the final set of answer choices for each MCQ.

# IMPLEMENTATION

## Answer Validation

- **MCQ Validation:** Compare the user's answer with the correct answer stored in the database.
- **Subjective Answer Validation:**

Semantic Score Calculation:

- Generate Embeddings
- Preprocess Text
- Compute Semantic Score
- Validation Criteria

# DATASET

- SQuAD- Stanford Question Answering Dataset
- T5 for extracting questions and answers from a PDF
- Fine-tune T5 on the SQuAD dataset to train the model
- PDF text used to generate questions and extract answers

# DATASET

| Column Name  | Description                        | Data Type       |
|--------------|------------------------------------|-----------------|
| id           | Unique question ID                 | Integer         |
| title        | Article title                      | String          |
| context      | Article text                       | String          |
| question     | Question text                      | String          |
| answers      | Answer(s) to the question          | List of Strings |
| answer_start | Starting character index of answer | Integer         |
| answer_end   | Ending character index of answer   | Integer         |

# DATASET

```
{
  "data": [
    {
      "paragraphs": [
        {
          "context": "The Normans (Norman: Nourmands; French: Normands; Latin: Normanni) were the people who in the 10th and 11th centuries gave their name to Normandy, a region in France. ",
          "qas": [
            {
              "answers": [
                {
                  "answer_start": 159,
                  "text": "France"
                }
              ],
              "id": "56ddde6b9a695914005b9628",
              "is_impossible": false,
              "question": "In what country is Normandy located?"
            }
          ]
        }
      ],
      "title": "Normans"
    }
  ],
  "version": 2
}
```

# TECHNOLOGY STACK

## SOFTWARE REQUIREMENTS

- Operating System: Windows 8 or above
- Programming language: Python 3.9
- IDE: Visual Studio Code, Google Colab
- Front-end: HTML, CSS, Python
- Back-end: Django
- Database: MySQL

# TECHNOLOGY STACK

## HARDWARE REQUIREMENTS

- Processor: 12th Gen Intel Core i7
- Processor Speed: 2.4GHz
- Memory: 32 GB RAM
- Graphics Card: NVIDIA RTX A4000
- Hard Disk Capacity: 1 TB



# CONCLUSION

- Automates question generation from PDF notes, improving efficiency and saving educator's time.
- Identifies at-risk students, enabling personalized interventions and support.
- Provides analysis of student performance with evaluation information.
- Future enhancements include attention tracking, sentiment analysis, adaptive learning modules, collaborative features, and gamification.
- Contributes to an inclusive learning environment, enhancing academic performance and supporting quality education.

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- [10] G. Kurdi, J. Leo, B. Parsia, U. Sattler, and S. Al-Emari, "A Systematic Review of Automatic Question Generation for Educational Purposes," *International Journal of Artificial Intelligence in Education*, vol. 30, pp. 121-147, 2019, doi: 10.1007/s40593-019-00186-y

The background of the slide features a large, faint watermark of the Mar Baselios Institute of Technology & Science logo. The logo is circular, with a laurel wreath border. Inside the circle, there is a gear-like border. The text "MAR BASELIOS INSTITUTE OF TECHNOLOGY & SCIENCE" is written in a pink arc at the top. In the center, there is a stylized blue bar chart with three bars of increasing height, and below it, a red and orange flame or flower-like shape. At the bottom, a blue banner contains the text "WISDOM CROWNS KNOWLEDGE" in white. The word "Mbits" is written in a large, grey, semi-transparent font across the center of the logo.

**THANK YOU**